

Introduction to Many-Body Quantum Information

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4 Bipartite entanglement (II)

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Main references for this section:

- Eric Chitambar and Gilad Gour, Rev. Mod. Phys. 91, 025001 (2019).
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- Martin B. Plenio, S. Virmani. Quant.Inf.Comput.7:1-51, (2007).

4.1 Quantum resources

A modern and conceptually precise way to understand quantum entanglement is through the framework of *quantum resource theories*.

Given a *quantum resource*, a resource theory is defined by specifying two fundamental ingredients:

- a set of *free states*: states that can be prepared without the quantum resource ;
- a set of *free operations*: operations that cannot generate quantum resource.

Consequently, states are not free states are *resource states*, and operations are not free operations are *resource operations*.

In the context of entanglement, i.e., by identifying entanglement as the quantum resource, we have

$$\text{Free states} \leftrightarrow \text{Seperable states} \quad (102)$$

$$\text{Entangled states} \leftrightarrow \text{Resource states} \quad (103)$$

$$\text{LOCC} \leftrightarrow \text{Free operations.} \quad (104)$$

This resource-theoretic formulation provides the foundation for quantifying entanglement via operational tasks such as state conversion, which we will discuss in the following sections.

4.2 The unit of entanglement resource

To quantify a resource, it is necessary to introduce a *standard unit* against which all states can be **compared**.

In the resource theory of bipartite entanglement, this role is conventionally played by the **maximally entangled Bell pair**,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle), \quad (105)$$

which represents the fundamental unit of entanglement. It serves as a **currency** of entanglement: arbitrary bipartite states can be compared via their interconvertibility with Bell pairs under LOCC.

4.3 Distillable entanglement and entanglement cost

In particular, for a bipartite state ρ , one considers the asymptotic transformations

$$\rho^{\otimes n} \xrightarrow{\text{LOCC}} |\Phi^+\rangle^{\otimes m}, \quad (106)$$

$$|\Phi^+\rangle^{\otimes m} \xrightarrow{\text{LOCC}} \rho^{\otimes n}, \quad (107)$$

in the limit $n \rightarrow \infty$, allowing for an arbitrarily small error.

The first process corresponds to *extracting* entanglement from a given state and leads to the notion of *distillable entanglement*. Intuitively, it quantifies how many Bell pairs can be obtained per copy of ρ .

Formally, the *distillable entanglement* $E_D(\rho)$ is defined as

$$E_D(\rho) := \sup \left\{ r : \lim_{n \rightarrow \infty} \inf_{\mathcal{E}_{\text{LOCC}}} \left\| \mathcal{E}_{\text{LOCC}}(\rho^{\otimes n}) - (\Phi^+)^{\otimes \lfloor rn \rfloor} \right\|_1 = 0 \right\}, \quad (108)$$

where $\mathcal{E}_{\text{LOCC}}$ is a trace-preserving LOCC map, and $\|\cdot\|_1$ denotes the *Schatten 1-norm*, defined for an operator A as

$$\|A\|_1 := \text{Tr} \sqrt{A^\dagger A}, \quad (109)$$

and $\Phi^+ := |\Phi^+\rangle\langle\Phi^+|$ is the density matrix corresponding to the maximally entangled Bell pair.

In other words, $E_D(\rho)$ is the optimal asymptotic rate r such that $\rho^{\otimes n}$ can be converted into approximately rn Bell pairs using LOCC.

Remark 4.1. To be more precise, one may write it as $E_D^{A:B}(\rho)$, with the superscript denoting the bipartite system ρ_{AB} . The Bell pairs obtained from the process are shared across the same bipartition.

Conversely, the second process corresponds to the *entanglement cost*, which quantifies how many Bell pairs are required to *create* a given state. It measures the minimal rate at which Bell pairs must be consumed to prepare ρ using LOCC. Formally, the *entanglement cost* $E_C(\rho)$ is defined as

$$E_C(\rho) := \inf \left\{ r : \lim_{n \rightarrow \infty} \inf_{\mathcal{E}_{\text{LOCC}}} \left\| \rho^{\otimes n} - \mathcal{E}_{\text{LOCC}} \left[(\Phi^+)^{\otimes \lfloor rn \rfloor} \right] \right\|_1 = 0 \right\}. \quad (\text{II0})$$

In other words, $E_C(\rho)$ is the minimal asymptotic rate r such that $\rho^{\otimes n}$ can be generated from approximately rn Bell pairs using LOCC.

Remark 4.2. When ρ is a pure state, these two quantities satisfy

$$E_D(\rho) = E_C(\rho) = S(\rho_A). \quad (\text{II1})$$

Remark 4.3 (Bound entanglement). For general mixed states,

$$E_D(\rho) < E_C(\rho). \quad (\text{II2})$$

The strict inequality reflects the intrinsic irreversibility of entanglement manipulation under LOCC. This is deeply consistent with the general irreversibility of open quantum systems.

Entangled states from which no Bell pairs can be distilled by LOCC are called *bound entangled states*.

Without proof, we state that both $E_D(\rho)$ and $E_C(\rho)$ are entanglement monotones for general mixed states. However, these quantities are typically difficult to compute in practice. Therefore, we introduce several alternative entanglement measures that are relatively more tractable, as discussed below.

4.4 Entanglement of formation

The *entanglement of formation* quantifies the minimal average entanglement needed to prepare a mixed state from pure states. For a bipartite state ρ ,

$$E_F(\rho) := \inf_{\{p_i, |\psi_i\rangle\}} \sum_i p_i E(|\psi_i\rangle), \quad \rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|, \quad (\text{II3})$$

where the infimum is over all pure-state ensembles realizing ρ , and

$$E(|\psi\rangle) := S(\text{Tr}_B |\psi\rangle \langle \psi|) \quad (\text{II4})$$

is the entanglement entropy of a pure state.

The entanglement of formation E_F satisfies:

- **Faithfulness:** $E_F(\rho) = 0$ iff ρ is separable.

- **LOCC monotonicity:** E_F does not increase under LOCC.
- **Convexity:** $E_F(\sum_i p_i \rho_i) \leq \sum_i p_i E_F(\rho_i)$.
- **Pure states:** for $\rho = |\psi\rangle\langle\psi|$, $E_F(|\psi\rangle) = S(\text{Tr}_B |\psi\rangle\langle\psi|)$.

Remark 4.4. By introducing the *regularized entanglement cost*

$$E_F^\infty(\rho) := \lim_{n \rightarrow \infty} \frac{1}{n} E_F(\rho^{\otimes n}), \quad (\text{II5})$$

one can show that

$$E_D(\rho) \leq E_C(\rho) = E_F^\infty(\rho) \leq E_F(\rho). \quad (\text{II6})$$

4.5 Concurrence

The *concurrence* is an entanglement measure that provides a computable characterization of bipartite entanglement, particularly for two-qubit systems. It plays a central role due to its direct relation to the entanglement of formation.

For a bipartite pure state $|\psi\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^2$, the concurrence is defined as

$$C(|\psi\rangle) := \sqrt{2(1 - \text{Tr} \rho_A^2)}, \quad \rho_A = \text{Tr}_B |\psi\rangle\langle\psi|. \quad (\text{II7})$$

Equivalently, it can be written as

$$C(|\psi\rangle) = |\langle\psi| (Y \otimes Y) |\psi^*\rangle|, \quad (\text{II8})$$

where $|\psi^*\rangle$ denotes complex conjugation in the computational basis.

For a general two-qubit mixed state ρ , the concurrence is defined via the *convex roof* construction,

$$C(\rho) := \inf_{\{p_i, |\psi_i\rangle\}} \sum_i p_i C(|\psi_i\rangle), \quad \rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|. \quad (\text{II9})$$

Remarkably, for two qubits this admits a closed-form expression (*Wootters formula*):

$$C(\rho) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}, \quad (\text{II10})$$

where $\{\lambda_i\}$ are the square roots (in decreasing order) of the eigenvalues of

$$\rho (Y \otimes Y) \rho^* (Y \otimes Y). \quad (\text{II11})$$

Remark 4.5. For two-qubit systems, the entanglement of formation is a monotonic function of the concurrence. Specifically, one has

$$E_F(\rho) = s\left(\frac{1 + \sqrt{1 - C(\rho)^2}}{2}\right), \quad (122)$$

where

$$s(x) = -x \log_2 x - (1 - x) \log_2 (1 - x) \quad (123)$$

is the binary entropy function.

- **Faithfulness:** $C(\rho) = 0$ iff ρ is separable (for two qubits).
- **Range:** $0 \leq C(\rho) \leq 1$.
- **LOCC monotonicity:** C does not increase under LOCC.
- **Pure states:** for Schmidt coefficients $\{\lambda_1, \lambda_2\}$: $C(|\psi\rangle) = 2\sqrt{\lambda_1 \lambda_2}$.

Remark 4.6. The concurrence is one of the few entanglement measures that can be computed exactly for general mixed states (in the two-qubit case). However, its extension to higher-dimensional systems does not admit a simple closed form and typically requires optimization via the convex roof construction.

4.6 Entanglement negativity

4.6.1 PPT criterion

The *positive partial transpose (PPT) criterion* provides a fundamental **necessary** condition for separability. For a bipartite state ρ , consider its partial transpose ρ^{T_B} . If ρ is separable, then

$$\rho^{T_B} \geq 0. \quad (124)$$

That is, the partial transpose of any separable state remains a positive semidefinite operator. Consequently, if ρ^{T_B} has at least one negative eigenvalue, then ρ is necessarily entangled.

Remark 4.7. If $\rho^{T_B} \geq 0$, the state is called *PPT*. However, PPT does not imply separability in general. It is both necessary and sufficient only for system dimensions of

$$\mathbb{C}^2 \otimes \mathbb{C}^2 \quad \text{and} \quad \mathbb{C}^2 \otimes \mathbb{C}^3. \quad (125)$$

The entangled states that are PPT corresponds to *bound entangled states*.

There exist entangled states that are PPT, i.e., $\rho^{T_B} \geq 0$ but ρ is not separable. These are known as *PPT (or bound) entangled states*. Such states have vanishing negativity and cannot be distilled into Bell pairs under LOCC.

4.6.2 Negativity and logarithmic negativity

The *entanglement negativity* provides a computable measure of bipartite entanglement based on the **partial transpose**. It is particularly useful for mixed states, as it avoids convex-roof optimizations.

The *negativity* is defined as

$$\mathcal{N}(\rho) := \frac{\|\rho^{T_B}\|_1 - 1}{2}, \quad (126)$$

where $\|\cdot\|_1$ is the trace norm. Equivalently, $\mathcal{N}(\rho)$ is the sum of the absolute values of the negative eigenvalues of ρ^{T_B} .

It is often convenient to consider the *logarithmic negativity*,

$$E_N(\rho) := \log \|\rho^{T_B}\|_1 = \log(2\mathcal{N}(\rho) + 1). \quad (127)$$

E_N satisfies

- **Partial faithfulness:** $\mathcal{N}(\rho) = 0$ for all separable states. However, there exist entangled states with $\mathcal{N}(\rho) = 0$ (PPT bound entangled states).
- **LOCC monotonicity:** $E_N(\rho)$ does not increase under LOCC.
- **Additivity:** $E_N(\rho \otimes \sigma) = E_N(\rho) + E_N(\sigma)$.
- **Computability:** both $\mathcal{N}$ and E_N can be computed efficiently via diagonalization.

Remark 4.8. The logarithmic negativity provides an upper bound on the distillable entanglement,

$$E_D(\rho) \leq E_N(\rho). \quad (128)$$

For pure states $|\psi\rangle$ with Schmidt coefficients $\{\lambda_i\}$,

$$E_N(|\psi\rangle) = \log \left(\sum_i \sqrt{\lambda_i} \right)^2 = 2 \log \left(\sum_i \sqrt{\lambda_i} \right), \quad (129)$$

which corresponds to the Rényi entropy of order $1/2$.

While E_N is not a complete entanglement measure (due to PPT-entangled states), its computational simplicity makes it a widely used diagnostic in many-body and quantum information settings.